Annabel Large

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Seeking research roles in machine learning and artificial intelligence for biological sequences

PhD graduation: December 2026

EDUCATION

University of California, Berkeley, Berkeley, CA **2020 - Present**
University of California, San Francisco, San Francisco, CA
PhD Thesis Advisor: Prof. Ian H. Holmes
UC Berkeley - UCSF Joint PhD Program in Bioengineering

Relevant UC Berkeley Coursework:
CS 289A: Introduction to Machine Learning
CS 288: Natural Language Processing

University of Tennessee at Knoxville (UTK), Knoxville, TN **2014 - 2019**
B.S., Chemical Engineering

SKILLS

Languages: Python (Jax, Pytorch for deep learning), Unix/bash scripting, Claude code

Domain knowledge: probabilistic modeling of biological sequences, deep learning (transformers, state space models, neural transducers), molecular evolution, statistical phylogenetics, bioinformatics, hidden markov models, protein evolution

EMPLOYMENT

University of California, Berkeley, Berkeley, CA **June 2021 - Present**
PhD Student, UC Berkeley - UCSF Graduate Program in Bioengineering
Expanding Protein Indel Models with Classical Mixtures and Deep Learning
Area of expertise: Mathematical modeling of biological sequence evolution through time
PhD Thesis Advisor: Prof. Ian H. Holmes
Department of Bioengineering

University of California, Berkeley, Berkeley, CA **Aug 2020-June 2021**
PhD Rotations, UC Berkeley - UCSF Graduate Program in Bioengineering
Rotation 1: Prof. Christoph Kirst, *Computational Theoretical Neuroscience*
Rotation 2: Prof. Priya Moorjani, *Causal Variant Discovery for South Asian Populations*
Rotation 3: Prof. Ian H. Holmes, *Machine learning for Directed Evolution of Proteins*

Swedish University of Agricultural Sciences, Alnarp, Sweden **Aug 2019-May 2020**
Swedish Fulbright Research Student, Bioinformatics
Network-based Multi-omic Data Integration for Exploration of Potato Plant Immunity Pathways

CITATIONS

Papers, Conference submissions:
LARGE, A., Holmes, I. "Nested birth-death processes are competitive with neural networks as time-dependent models of protein evolution." ICML 2026, main track.

Holmes, I., **LARGE, A.** “Closed-form Baum-Welch for TKF-based evolutionary models.” NeurIPS 2026 workshop: AI for Stochastic Dynamics.

Holmes, I., **LARGE, A.** “Variational inference in coupled models of amino acid substitution.” NeurIPS 2026 workshop: AI for Stochastic Dynamics.

LARGE, A., Holmes, I. “Nested Birth-Death Processes and Neural Transducers as Models of Time-Dependent Protein Evolution.” NeurIPS 2026 workshop: Women in Machine Learning.

Resjö, S., Willforss, J., **LARGE, A.**, Siino, V., Alexandersson, E., Levander, F., Andreasson, E. “Comparative proteomic analyses of potato leaves from field-grown plants grown under extremely long days.” Plant Physiology and Biochemistry (2024).

Olayide, P., **LARGE, A.**, Stridh, L., Rabbi, I., Baldermann, S., Stavolone, L., Alexandersson, E. “Gene Expression and Metabolite Profiling of Thirteen Nigerian Cassava Landraces to Elucidate Starch and Carotenoid Composition.” Agronomy (2020)

Kelley, B., Ellis, J., **LARGE, A.**, Schneider, L., Jacobson, D., Johnson, J. "Whole-Genome Sequencing and Bioinformatic Analysis of Environmental, Agricultural, and Human Campylobacter jejuni Isolates From East Tennessee." Frontiers in Microbiology 11 (2020)

Furches, A., Kainer, D., Weighill, D., **LARGE, A.**, …, Jacobson, D. “Finding cell wall regulatory mechanisms through multiple lines of evidence.” Frontiers in Plant Sciences (2019).

Invited Talks:

LARGE, A., Holmes, I. “Nested birth-death processes are competitive with parameter-heavy neural networks as time-dependent models of protein evolution.”

- Probabilistic Modeling in Genomics 2026, Berkeley, CA, USA (2026).
- Mathematical and Computational Evolutionary Biology, Crete, Greece (2026).
- The Annual Meeting of the Society for Molecular Biology and Evolution, Copenhagen, Denmark (2026).

LARGE, A., Holmes, I. “Fitting Indel Evolution Model Parameters to PFam Protein Alignments with a GPU-accelerated Framework.” LEGEND 2024, Crete, Greece (2024).
LARGE, A., Weighill, D., Joubert, W., Furches, A., Kainer, D., Jones, P., Podar, M., Jacobson, D. “Network-based Approaches in the Analyses of the Oral Microbiome.” INFORMS Computing Society Conference, Knoxville, TN (2019).

AWARDS

Hearts to Humanities Fellowship	2024
NVidia Hardware Grant Recipient (two A6000 GPUs)	2022
UC Berkeley Bioengineering Brodie Endowed Fellowship	2022
National Science Foundation Graduate Research Fellowship Program; Honorable Mentions	2021, 2019
Fulbright Research Scholar; Sweden	2019

TEACHING

BioEng 10: Introduction to Bioengineering **Aug-Dec 2024**
University of California, Berkeley
Head Instructor: Prof. Iain Clark
Role: Graduate Student Instructor

- Taught 100 undergraduate students across 3 lectures per week
- Co-created lecture materials and practice problems for all 300 undergraduate students across all BioEng 10 classes
- Created midterm assignment, responsible for 10% of final course grade

Center for Computational Biology: Python bootcamp **2022, 2024**
University of California, Berkeley
Role: Teaching Assistant

- Assisted undergraduate students, graduate students, and industry partners in python bootcamp assignments, spanning one week of material
- Taught at 4 bootcamps across 2 years

Accelerating climate resilient plant breeding by applying –omics and artificial intelligence **April 20-24, 2020**
Swedish University of Agricultural Sciences, Alnarp, Sweden
Head Instructor: Prof. Erik Alexandersson
Role: Guest Instructor

- Created hands-on software demonstration and corresponding student assignment for identifying differentially expressed genes (in R programming language)

SERVICE

BioE Admissions Committee **Dec 20xx, 20xx**
Selecting the next class of PhD students for the UC Berkeley - UCSF Graduate Program in Bioengineering

- Screen ~xx applicants per cycle
- Collaborate across departments and schools to select best candidates

BioE Retreat Committee Co-Chair **Jan 2022-Jan 2023**
Co-lead organization of the 2022 BioE Department Retreat, the first in-person retreat after COVID pandemic

- Created programming for +250 students, staff, and faculty across 3 days
- Personally responsible for registration, transportation, and catering
- In coordination with the Diversity, Equity, Inclusion, and Belonging (DEIB) Committee and Faculty Executive Committee, initiated a new “State of DEIB in Bioengineering” talk